A Digit-Avoidance Conjecture for Powers of Two

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Abstract

We investigate a digit-avoidance property of the powers of two in base ten. The number

$$2^{16} = 65536$$

contains none of the decimal digits $1, 2, 4, 8$. We conjecture that it is the only nonnegative integral power of two with this property.

Elementary modular arguments impose necessary restrictions on any possible solution. We establish a periodicity result for decimal suffixes and describe a computational method for enumerating exponent residue classes that satisfy increasingly long suffix restrictions. Two complementary Python scripts provide a framework for searching individual powers and studying the associated residue classes.

The uniqueness assertion remains open. Computational evidence over finite ranges and patterns in residue-class counts are not presented as proofs of the conjecture.

1 Introduction

The decimal expansion of an integer can exhibit properties that are not immediately apparent from its algebraic representation. For example,

$$2^{16} = 65536$$

is a power of two whose decimal digits belong entirely to

$$\{0, 3, 5, 6, 7, 9\}.$$

In particular, none of the digits $1, 2, 4, 8$ occurs.

This observation motivates a simple question: is 65536 the only power of two whose decimal representation avoids all four of these digits?

Although the question is elementary to state, proving uniqueness requires controlling decimal digits over arbitrarily many positions. The elementary results below restrict the possible exponents. We then establish a modular framework for studying longer decimal suffixes and describe computational experiments motivated by it.

2 Definitions and Main Conjecture

Let

$$\mathcal{D} = \{1, 2, 4, 8\}.$$

Definition 2.1. For $n \in \mathbb{Z}_{\geq 0}$, let $P(n)$ denote the statement that the standard decimal representation of 2^n contains no digit belonging to $\mathcal{D}$.

The conjectured example is

$$2^{16} = 65536,$$

so $P(16)$ holds.

Conjecture 2.2 (Uniqueness conjecture). For every $n \in \mathbb{Z}_{\geq 0}$,

$$P(n) \iff n = 16.$$

The case $n = 0$ is not a solution, since $2^0 = 1$. The case $n = 16$ establishes existence. The remaining task is to prove that no other exponent satisfies $P(n)$.

3 Restrictions from the Final Decimal Digit

For $n \geq 1$, the final decimal digit of 2^n follows the periodic cycle

$$2,\ 4,\ 8,\ 6.$$

Equivalently,

$$2^n \bmod 10 = \begin{cases} 2, & n \equiv 1 \pmod 4, \\ 4, & n \equiv 2 \pmod 4, \\ 8, & n \equiv 3 \pmod 4, \\ 6, & n \equiv 0 \pmod 4. \end{cases}$$

Since $2, 4, 8$ are forbidden, any positive solution must end in the digit 6.

Proposition 3.1. *If $n \geq 1$ and $P(n)$ holds, then*

$$n \equiv 0 \pmod 4.$$

Proof. The final digit of 2^n must be 6. The final-digit cycle shows that this occurs precisely when $n \equiv 0 \pmod 4$. □

Consequently, every positive candidate exponent is a multiple of four. Writing $n = 4m$, we obtain

$$2^n = 2^{4m} = 16^m.$$

Thus the uniqueness conjecture is equivalent to the assertion that 16^m contains a forbidden decimal digit for every positive integer $m \neq 4$.

4 Restrictions from the Final Two Decimal Digits

The powers of 16 modulo 100 satisfy

$$\begin{aligned} 16^1 &\equiv 16 \pmod{100}, \\ 16^2 &\equiv 56 \pmod{100}, \\ 16^3 &\equiv 96 \pmod{100}, \\ 16^4 &\equiv 36 \pmod{100}, \\ 16^5 &\equiv 76 \pmod{100}, \\ 16^6 &\equiv 16 \pmod{100}. \end{aligned}$$

Hence the final two digits cycle through

$$16,\ 56,\ 96,\ 36,\ 76$$

with period five.

When $m \equiv 1 \pmod 5$, the final two digits of 16^m are 16, which contains the forbidden digit 1. Therefore, a solution must satisfy $m \not\equiv 1 \pmod 5$.

Proposition 4.1. *If $n \geq 1$ and $P(n)$ holds, then*

$$n \equiv 0, 8, 12, \text{ or } 16 \pmod{20}.$$

Proof. The preceding proposition gives $n = 4m$. The final-two-digit cycle excludes $m \equiv 1 \pmod 5$, leaving

$$m \equiv 0, 2, 3, \text{ or } 4 \pmod 5.$$

Multiplying these congruences by four yields the result. □

Remark 4.2. These restrictions are necessary, not sufficient. Avoiding forbidden digits in the final two positions does not imply that the other digits are also allowed.

5 Periodicity of Decimal Suffixes

We now consider the last k decimal digits of 2^n. For a positive integer k, define

$$P_k = 4 \cdot 5^{k-1}.$$

We first establish the relevant multiplicative-order formula.

Lemma 5.1. *For every integer $k \geq 1$,*

$$\operatorname{ord}_{5^k}(2) = 4 \cdot 5^{k-1}.$$

Proof. Since 2 has order 4 modulo 5, any exponent r satisfying $2^r \equiv 1 \pmod{5^k}$ must be divisible by four. Write $r = 4m$.

For $m \geq 1$, the standard lifting-the-exponent formula gives

$$v_5(16^m - 1) = v_5(16 - 1) + v_5(m) = 1 + v_5(m),$$

where $v_5(x)$ denotes the exponent of 5 in the prime factorization of x.

Since $2^{4m} = 16^m$, the congruence

$$2^{4m} \equiv 1 \pmod{5^k}$$

holds precisely when $1 + v_5(m) \geq k$, or equivalently when $5^{k-1} \mid m$.

The smallest positive such exponent is therefore $4 \cdot 5^{k-1}$, as required. □

Theorem 5.2 (Periodicity of the last k digits)**.** *For every $k \geq 1$ and every $n \geq k$,*

$$2^{n+P_k} \equiv 2^n \pmod{10^k}, \qquad P_k = 4 \cdot 5^{k-1}.$$

Consequently, the last k decimal digits of 2^n depend only on $n \bmod P_k$, once $n \geq k$.

Proof. Because $n \geq k$,

$$2^n \equiv 0 \pmod{2^k}.$$

The same is true of 2^{n+P_k}. Thus

$$2^{n+P_k} \equiv 2^n \pmod{2^k}.$$

By the lemma,

$$2^{P_k} \equiv 1 \pmod{5^k}.$$

Multiplying by 2^n gives

$$2^{n+P_k} \equiv 2^n \pmod{5^k}.$$

Since 2^k and 5^k are coprime, the Chinese remainder theorem yields

$$2^{n+P_k} \equiv 2^n \pmod{10^k}.$$

Repeated application establishes the claimed dependence on the exponent modulo P_k. □

Remark 5.3. The integer 2 is not invertible modulo 10^k. The periodicity result therefore does not follow by treating 2 as an element of the multiplicative group modulo 10^k. Instead, the proof uses the vanishing of 2^n modulo 2^k for $n \geq k$, together with the multiplicative order of 2 modulo 5^k.

Remark 5.4. The theorem establishes a period; it does not assert that P_k is the smallest possible period of the suffix sequence. When interpreting suffixes shorter than the full decimal representation, leading zeros may be used to pad the suffix to length k. Such padding introduces no forbidden digits.

6 Admissible Residue Classes

The periodicity theorem allows the suffix restriction to be studied through a finite set of exponent residue classes.

Definition 6.1. For $k \geq 1$, let S_k be the set of residue classes modulo $P_k = 4 \cdot 5^{k-1}$ whose representatives $n \geq k$ have last k decimal digits containing no digit from $\mathcal{D}$. The periodicity theorem ensures that this property is independent of the choice of representative, provided the representative is at least k.

Write

$$s_k = |S_k|.$$

There are P_k residue classes in total, so

$$0 \leq s_k \leq P_k.$$

Any exponent satisfying the original digit-avoidance property must belong to an admissible suffix class for every suffix length $k \leq n$. However, admissibility for a fixed suffix length does not establish that the entire decimal expansion is admissible.

The sequence of class counts s_k provides a way to investigate how restrictive the suffix conditions become as k increases.

7 A Conjectured Residue-Class Counting Pattern

Computational experiments using `trailing_classes.py` are designed to count the admissible classes as k increases. A pattern suggested by the preliminary analysis is

$$s_1 = 1, \qquad s_k = 4 \cdot 3^{k-2} \quad (k \geq 2).$$

This formula is recorded as a conjecture about the class counts, not as a theorem. A complete proof would require a rigorous description of how admissible classes at level k lift to classes at level $k + 1$.

In particular, the fact that each class has five possible extensions does not by itself imply that exactly three extensions remain admissible. The new decimal digit depends on the extension, and the number of allowed digits depends on whether the digit is odd or even. Any proposed recurrence must account for this structure.

The script provides a finite computational test of the proposed pattern for the values of k it examines. Its output should be reported alongside the precise run parameters when presenting numerical evidence. Such evidence does not substitute for a proof of the formula for all k.

Even if the proposed formula were established, it would describe admissible suffix classes rather than directly count powers of two whose entire decimal expansions avoid the forbidden digits. The distinction is essential: a number can pass a suffix test and still contain a forbidden digit farther to the left.

8 Computational Methods

Two complementary Python scripts accompany this investigation. The source files are available in the repository:

- `scripts/pow2_digits.py`: exact search and suffix sieve.
- `scripts/trailing_classes.py`: residue-class enumeration.

8.1 Exact search with suffix filtering

The script `pow2_digits.py` implements two methods.

1. **Direct search.** For each exponent in a specified finite interval, the program constructs 2^n exactly and checks its full decimal representation.
2. **Suffix sieve followed by exact verification.** The program tracks $2^n \bmod 10^K$ incrementally. If the resulting K-digit suffix contains a forbidden digit, the exponent is rejected. Every candidate that passes the suffix test is then checked against the full integer 2^n.

The sieve is logically exact. If a forbidden digit occurs among the last K digits, the full decimal representation cannot satisfy the conjectured property. Passing the sieve, however, is only a necessary condition, so the full check remains essential.

The program compares the direct and sieved methods on a common range and asserts that their reported solutions agree there. This cross-check helps detect implementation errors, although it does not prove anything about exponents outside the searched range.

8.2 Residue-class enumeration

The script `trailing_classes.py` enumerates exponent residue classes modulo P_k and retains classes whose last k digits avoid the forbidden set. It extends the classification from one suffix length to the next by examining the possible extensions of each surviving class.

This computation investigates the structure of the suffix restriction rather than checking only individual powers. The number of surviving classes at each level can be compared with the proposed counting pattern in the preceding section.

Both scripts are finite computations. Their conclusions must be limited to the exponent ranges and suffix lengths actually examined.

8.3 Reproducibility

The scripts can be run with Python 3 using commands of the form

```
python pow2_digits.py 20000 2000000
python trailing_classes.py 13
```

The first command sets the direct-search limit to 20,000 and the sieved-search limit to 2,000,000. The second requests residue-class counts through suffix length 13. These are example commands, not a claim that a particular execution has been performed or independently verified here.

For a reproducible report, the repository should preserve the script versions, the exact command-line arguments, the Python version, and the resulting output. Any numerical claims in the paper should match that recorded output.

Remark 8.1. A finite search can establish

$$P(n) \implies n = 16 \qquad \text{for } 0 \le n \le N$$

for a specified bound N, provided every exponent in that range is checked correctly. It cannot establish the same statement for all nonnegative integers n.

9 Further Questions

The elementary restrictions leave infinitely many candidate exponents. A proof of uniqueness would need to control the decimal digits across arbitrarily many positions or introduce another argument that excludes every exponent other than 16.

The modular approach suggests several questions:

1. Can the admissible suffix classes be described by an exact recurrence as the suffix length increases?
2. Is the observed formula $s_k = 4 \cdot 3^{k-2}$ valid for every $k \geq 2$, and if so, can it be proved?
3. Can bounds on the number of admissible suffix classes yield meaningful restrictions on full powers of two?
4. Can any such counting argument establish uniqueness, or would it only show that admissible exponents are sparse?
5. Is there a useful connection with established results on the distribution of digits in sequences such as 2^n?

A density estimate, even if proved, would not alone establish uniqueness. A set can have density zero and still contain infinitely many elements. The central challenge remains to exclude every exponent other than 16.

10 Conclusion

The identity

$$2^{16} = 65536$$

provides an example of a power of two whose decimal expansion avoids the digits $1, 2, 4, 8$.

Elementary modular arguments show that any positive solution must have exponent divisible by four. An analysis of the final two digits further restricts the exponent to the residue classes

$$n \equiv 0, 8, 12, \text{ or } 16 \pmod{20}.$$

The periodicity theorem provides a rigorous foundation for studying longer decimal suffixes through finite residue-class enumerations. The accompanying scripts offer reproducible methods for exploring both individual powers and suffix-class counts.

Nevertheless, neither finite computation nor the observed counting pattern establishes uniqueness over all exponents. The central conjecture remains open:

65536 is the unique nonnegative power of two avoiding the decimal digits $1, 2, 4, 8$.